

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool : Class Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.
2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

I K PHANI SRIDHAR YOGI 192311091 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K-P-Sridhar

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

1. Fundamental Concepts Used in Web Development

1. XML – Extensible Markup Language

XML is a markup language used to store, organize, and exchange structured data. It allows developers to create their own tags.

Example:

XML

```
<student>
  <name>Manisha</name>
  <department>CSE</department>
  <marks>89</marks>
</student>
```

Purpose: XML separates data from presentation and is commonly used for data exchange between different systems.

2. JSP – JavaServer Pages

JSP is a server-side Java technology used to create dynamic web pages. It allows HTML pages to interact with Java code and backend data.

Purpose: JSP generates dynamic HTML content based on user requests, database information, or application logic.

Example: A student portal can use JSP to display a student's marks retrieved from a database.

3. MVC Architecture – Model View Controller

MVC is a software architectural pattern that divides an application into three components:

- Model: Manages data and business logic.
- View: Displays information to the user.
- Controller: Receives user requests and controls the flow between Model and View.

Flow:

User → Controller → Model → Database → Controller → View → User

Purpose: MVC provides separation of concerns, making applications easier to develop, test, and maintain.

4. XSLT – Extensible Stylesheet Language Transformations

XSLT is a language used to transform XML documents into other formats, such as HTML, XML, or text.

Example:

XML Data + XSLT → HTML Web Page

Purpose: It separates XML data from its presentation and is especially useful for generating web reports from XML data.

5. XML Schema – XSD

XML Schema, commonly called XSD (XML Schema Definition), defines the structure and rules of an XML document.

It can specify:

- Required elements
- Data types
- Number of occurrences
- Allowed values
- Restrictions on data

Purpose: XSD validates whether an XML document follows the required structure and data rules.

6. JSTL – JSP Standard Tag Library

JSTL is a collection of standard tags used in JSP pages. It reduces the need to write Java code directly inside JSP.

Common JSTL functions include:

- Conditional statements
- Loops
- Formatting
- Database-related operations
- XML processing

Purpose: JSTL makes JSP pages cleaner, easier to read, and easier to maintain.

7. PHP – Hypertext Preprocessor

PHP is a server-side scripting language used to develop dynamic web applications.

It can interact with:

- Databases
- Forms
- Sessions

- APIs
- Files

Example: An e-commerce website can use PHP to retrieve product information from a database and display it dynamically.

8. Database Connectivity

Database connectivity is the process of connecting a web application to a database so that it can store, retrieve, update, and delete information.

In Java applications, technologies such as JDBC can be used.

In PHP applications, PDO or database-specific extensions can be used.

Purpose: It allows applications to communicate with databases such as MySQL, PostgreSQL, or Oracle.

9. Parser

A parser analyzes data according to its defined syntax and converts it into a form that a program can process.

For XML, common parsing approaches include:

- DOM: Builds an in-memory tree of the complete XML document.
- SAX: Reads XML sequentially and generates events.

Purpose: A parser allows applications to read, analyze, validate, and process structured data.

Summary Table

Concept	Main Purpose
XML	Stores and exchanges structured data
JSP	Generates dynamic web pages using Java
MVC	Organizes application into Model, View, and Controller
XSLT	Transforms XML into HTML/XML/text
XML Schema (XSD)	Defines and validates XML structure
JSTL	Provides standard tags for JSP pages
PHP	Develops dynamic server-side web applications

Concept	Main Purpose
Database Connectivity	Connects applications with databases
Parser	Reads and processes structured data

2. Interconnection of XML, JSP, PHP, MVC, XSLT, Database Connectivity, and Parsers

These technologies can work together to create a data-driven web application. Each technology performs a different role in the application.

Overall Architecture

A simplified flow is:

User → Controller → Model/Business Logic → Database/XML →

Parser/Validation → JSP/XSLT → View → User

PHP can also be used as the server-side technology instead of Java/JSP.

Step 1: XML Stores Structured Data

XML represents data in a structured, hierarchical form.

For example:

XML

```
<student>
```

```
  <id>101</id>
```

```
  <name>Manisha</name>
```

```
  <marks>89</marks>
```

```
</student>
```

The tags describe the meaning of the data, making XML suitable for data exchange between applications.

Step 2: XML Schema Validates the Data

Before XML data is processed, it can be validated against an XML Schema (XSD).

For example, XSD can ensure that:

- id contains an integer.
- name is present.
- marks contains a valid numeric value.
- Required elements are not missing.

Therefore:

XML → XSD Validation → Valid XML

If the XML does not follow the schema, the application can reject it before further processing.

Step 3: Parser Processes XML

After receiving XML, an XML parser reads and analyzes it.

For example:

XML → DOM/SAX Parser → Application

DOM is useful when the complete XML structure needs to be accessed, while SAX is useful for processing large XML documents sequentially with low memory consumption.

Step 4: XSLT Transforms XML

When XML data needs to be presented as a web report, XSLT can transform it into HTML.

XML + XSLT → HTML

For example, student XML data can be transformed into an HTML table containing student names and marks.

Thus, XML handles the data, while XSLT handles its transformation/presentation.

Step 5: MVC Organizes the Application

MVC provides the overall organization of the web application.

Model

The Model handles:

- Data
- Business logic
- Database operations

View

The View displays information to users.

JSP can act as the View in a Java-based MVC application.

Controller

The Controller receives requests and coordinates the Model and View. In a JSP/Servlet application, a Servlet commonly acts as the Controller.

The flow becomes:

User → Servlet Controller → Model → Database → Model → JSP View → User

Step 6: JSP Performs Server-Side Processing

JSP generates dynamic web pages based on information received from the Controller and Model.

For example:

1. User requests student details.
2. Servlet receives the request.
3. Model retrieves student information.
4. JSP receives the data.
5. JSP generates HTML.
6. HTML is sent to the browser.

JSTL can be used within JSP to perform common operations such as iteration and conditional display without embedding large amounts of Java code in the page.

Step 7: PHP as an Alternative Server-Side Technology

PHP can perform a role similar to Java/JSP in server-side web development.

For example:

Browser → PHP Application → Database → PHP → HTML → Browser

PHP can:

- Receive user requests
- Process form data
- Access databases
- Apply business logic
- Generate dynamic HTML
- Communicate with external services

Modern PHP frameworks can also implement MVC architecture.

Step 8: Database Connectivity

A data-driven application needs a way to communicate with its database.

In Java:

JSP/Servlet → JDBC/JPA → Database

In PHP:

PHP → PDO/Database Library → Database

The database can store information such as:

- User details
- Product information
- Student records
- Orders
- Employee information

Database connectivity enables CRUD operations:

- Create
- Read
- Update
- Delete

Complete Interconnection

A typical Java-based application can work as follows:

User Request



Servlet / Controller



Model / Business Logic



JDBC / Database Connectivity



Database



Model



JSP / JSTL



HTML Response



User

When XML is involved:

XML → Parser → XSD Validation → Application Processing → XSLT Transformation → HTML

Therefore, the technologies have complementary roles:

Technology	Role in Web Application
XML	Stores/exchanges structured data
XSD	Validates XML
Parser	Reads and processes XML
XSLT	Transforms XML into presentable formats
MVC	Organizes the complete application

Technology	Role in Web Application
Servlet	Commonly acts as Controller in Java MVC
JSP	Commonly acts as View
JSTL	Simplifies JSP presentation logic
PHP	Alternative server-side technology
JDBC/PDO	Provides database connectivity
Database	Permanently stores application data

Conclusion

XML, XSD, parsers, XSLT, JSP, JSTL, MVC, PHP, and database connectivity are complementary technologies used at different stages of web application development. XML provides structured data, XSD validates it, parsers process it, XSLT transforms it, MVC organizes the application, JSP/PHP performs server-side processing, and database connectivity allows persistent data storage. Together, these technologies can be used to build structured, maintainable, and data-driven web applications.